

11. (a) Arsenic oxide, As_2O_3 , is prepared on an industrial scale by roasting arsenic-containing ores such as arsenopyrite, FeAsS , in air. The other products formed are iron(III) oxide and sulfur dioxide.

(i) State the oxidation state of arsenic in As_2O_3 . [1]

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(ii) Give a balanced chemical equation for the industrial production of As_2O_3 from FeAsS . [2]

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- (b) As_2O_3 is moderately soluble in water. 100 cm^3 of a saturated solution at 25°C contains 2.06 g .

When dissolved in water, the oxide reacts to form arsenous acid.



- (i) Calculate the concentration of the arsenous acid, in mol dm^{-3} , in the saturated solution. [3]

concentration of $\text{H}_3\text{AsO}_3 = \dots\dots\dots \text{mol dm}^{-3}$

- (ii) A solution of arsenous acid has a pH of 5.11.

Calculate the hydrogen ion concentration of this solution. [1]

$[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$



- (c) The formula for arsenous acid can be written as $\text{As}(\text{OH})_3$ since it contains three hydroxyl (OH) groups bonded to arsenic.

Suggest the shape around the arsenic atom in $\text{As}(\text{OH})_3$. Justify your answer by using VSEPR theory. [3]

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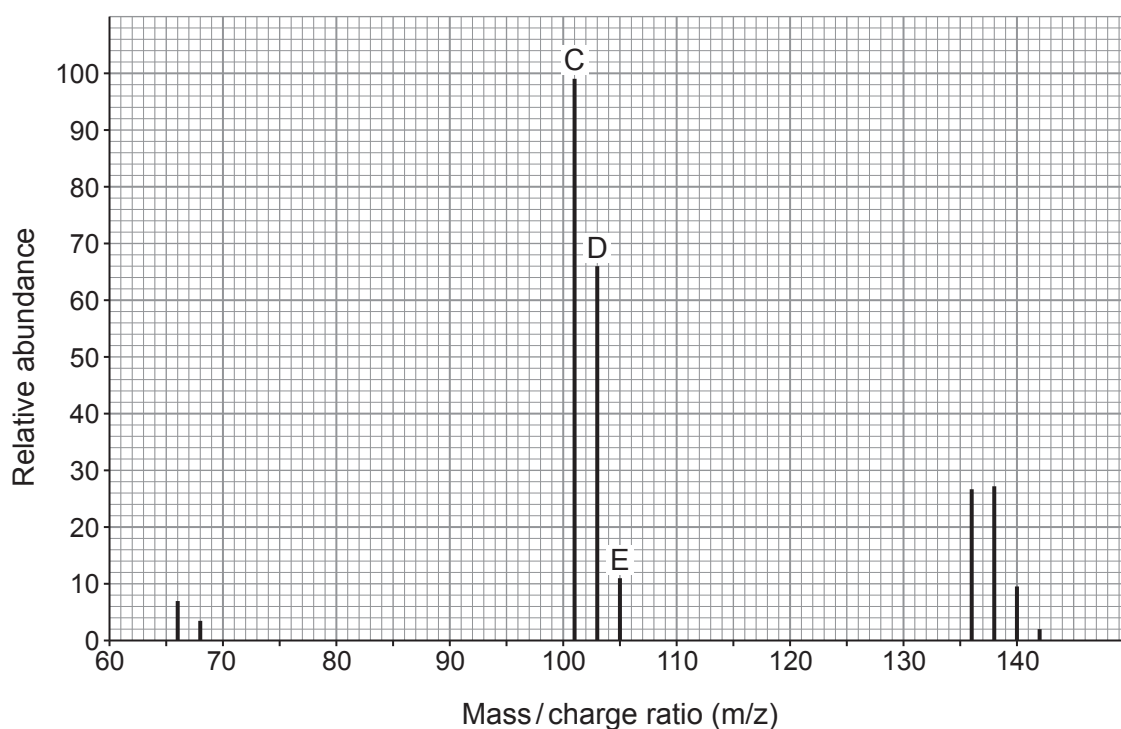
- (d) Phosphorus can form two chlorides, PCl_3 and PCl_5 .

0.181 g of a chloride of phosphorus gave 39 cm^3 of vapour at 1 atm pressure when heated to 87°C . The sample was completely vapourised.

Show that the chloride of phosphorus was PCl_3 . [4]



- (e) The molecular ion region of the mass spectrum of PCl_3 , is shown below.



- (i) Identify the species responsible for peak C at m/z 101.

[1]

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- (ii) Explain why the height ratio of peaks C:E is 9:1.

[2]

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